



Agritourism and Rural Livelihood Enhancement in Ghana: Evidence from a Mixed-Methods Case Study of Adjeikrom



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Abstract: Agritourism has increasingly been recognized as a strategy for livelihood diversification within sustainable rural development; however, robust empirical evidence regarding its household-level socioeconomic contributions in sub-Saharan Africa remains limited. This study investigates the extent to which agritourism contributes to rural livelihood enhancement in Adjeikrom, Ghana. A pragmatic research paradigm was adopted, and a convergent mixed-methods case study design was employed. Primary data were collected from 100 households using structured questionnaires, complemented by in-depth interviews, focus group discussions, and stakeholder consultations. Quantitative data were analyzed using descriptive statistics and Pearson correlation analysis in IBM SPSS Statistics, while qualitative data were analyzed thematically. Findings indicate that Adjeikrom possesses diverse agritourism resources, including cocoa farm tours, village experiences, and mountain hiking, widely recognized and positively evaluated by residents. Direct livelihood benefits were reported by 74% of households through employment and enterprise development. Correlation analysis revealed a weak, non-significant positive association between household income and perceived livelihood enhancement ($r = 0.250$, $p > 0.05$), suggesting agritourism contributes to household well-being without being the principal determinant of livelihood outcomes. Qualitative evidence showed that agricultural productivity was strengthened through technical training, improved input access, and institutional partnerships linked to agritourism, alongside indirect benefits such as improved infrastructure and social capital. By integrating household perceptions, livelihood outcomes, and agricultural support mechanisms within a single analytical framework, the study provides a more comprehensive understanding of agritourism's multidimensional role in rural development. Findings suggest agritourism should be regarded as an emerging, transitional livelihood pathway, with greater impact achievable through stronger infrastructure, institutional coordination, and community participation linking agriculture, tourism, and local economic development.

Keywords: Adjeikrom; Agritourism; Ghana; Livelihood diversification; Rural livelihood enhancement; Sustainable livelihood enhancement

1. Introduction

Agritourism is widely recognized in development literature as a sustainable rural development strategy that integrates agriculture and tourism to diversify rural economies, generate employment, and improve household incomes (Pehin Dato Musa & Chin, 2022). Wu et al. (2024) posited that agritourism functions as a complementary livelihood pathway that enables rural households to reduce dependence on subsistence farming while benefiting from value-added services such as farm tours, cultural experiences, local product sales and hospitality services. When effectively developed, it strengthens rural value chains, promotes local entrepreneurship, enhances community participation in tourism and contributes to broader socio-economic development. In this sense, agritourism is expected to serve as a viable mechanism for improving rural livelihoods, particularly in agriculturally based economies such as Ghana.

In Ghana, however, agritourism remains at a relatively early and underdeveloped stage, with its potential not

fully harnessed for rural livelihood transformation (Adom et al., 2021). Although pilot initiatives such as the agritourism project in Adjeikrom in the Eastern Region have demonstrated the feasibility of linking agriculture with tourism, many rural communities continue to experience limited participation in agritourism activities and remain heavily dependent on traditional agriculture, which is often characterized by low productivity, seasonal income instability, and vulnerability to environmental and market shocks. Furthermore, Kolawole et al. (2023) opined that agritourism development in most rural areas is constrained by inadequate infrastructure, weak institutional support, limited awareness, and insufficient community engagement, thereby limiting its capacity to significantly influence income diversification and employment creation at the household level.

Despite the growing policy and academic interest in agritourism as a rural development tool, there remains a lack of empirical, context-specific evidence on its actual contribution to rural livelihood enhancement in Ghana, particularly at the household level (Adu-Ampong & Kimbu, 2020). Existing studies tend to be fragmented, often focusing broadly on rural tourism or agricultural development without isolating agritourism as a distinct livelihood strategy (Ammirato et al., 2020; Kolawole et al., 2023). Additionally, Saepudin et al. (2022) posited that limited attention has been given to understanding the extent of household participation in agritourism, its impact on income and employment, and the perceptions and challenges experienced by rural communities engaged in such activities. This creates a clear research gap that necessitates an in-depth, mixed-methods investigation to assess how agritourism contributes to livelihood outcomes in Adjeikrom, thereby informing both policy and practice for sustainable rural development in Ghana.

2. Literature Review

2.1 Agritourism Development in Rural Development Contexts

Tourism is generally recognized as a significant driver of structural transformation in developing economies, particularly in contexts where rural livelihoods remain heavily dependent on primary agriculture (Ghadami et al., 2022). In many Sub-Saharan African countries, including Ghana, rural economies are characterized by low income levels, seasonal agricultural employment, limited access to diversified income sources, and vulnerability to climate and market shocks. Within this context, tourism is increasingly positioned as a complementary sector capable of stimulating rural economic diversification, enhancing local enterprise development, and improving household welfare (Ma et al., 2024). In Ghana, national development frameworks such as medium-term development plans and poverty reduction strategies have progressively emphasized the integration of tourism and agriculture as a deliberate strategy for rural transformation (Adu-Ampong & Kimbu, 2020). This policy direction reflects a growing recognition that agriculture alone is often insufficient to sustain rural livelihoods, particularly in areas where productivity is constrained by limited technology adoption, fluctuating commodity prices and environmental stressors. Consequently, agritourism is promoted as a strategic linkage between the agricultural sector and the tourism industry, enabling rural communities to move beyond subsistence-oriented production toward more diversified and market-oriented livelihood systems (Unjia et al., 2024).

Within this policy orientation, agritourism has emerged as a hybrid development approach that combines primary agricultural production with tourism-related services (Kothari & Perwej, 2021; Streifeneder et al., 2023). It enables rural households and communities to leverage existing agricultural resources such as farms, landscapes, crops, livestock, and cultural farming practices as tourism assets. Through activities such as farm tours, harvest participation, cultural demonstrations, and on-farm product sales, agritourism creates opportunities for rural producers to engage directly with visitors and capture additional value from their agricultural activities (Pehin Dato Musa & Chin, 2022; Yadav & Kumari, 2024). This integration strengthens rural participation in broader local and global value chains, while simultaneously enhancing the visibility and marketability of rural products. From a rural development perspective, agritourism is increasingly conceptualized as a livelihood diversification strategy aimed at reducing overdependence on primary agriculture while expanding rural income portfolios (Zvavahera & Chigora, 2023). According to Bannor et al. (2022), diversification is particularly important in rural economies where agricultural incomes are often unstable due to seasonal production cycles, price volatility and climate variability. By introducing non-farm income opportunities, agritourism contributes to income smoothing, risk reduction, and enhanced household resilience (Zvavahera & Chigora, 2023). It also fosters entrepreneurial activities within rural communities, encouraging the emergence of micro and small enterprises linked to hospitality, food services, transportation, and local craft production.

This perspective aligns with broader global rural development thinking that emphasizes the importance of multi-sectoral rural economies capable of generating employment, improving living standards, and strengthening adaptive capacity. In this regard, Zvavahera & Chigora (2023) posited that agritourism is not merely an economic activity but also a spatial development strategy that reconfigures rural areas into multifunctional socio-economic landscapes. Rural spaces are transformed from purely production-oriented environments into integrated spaces of production, consumption, learning, and cultural exchange (Neilson, 2022). This multifunctionality enhances the developmental value of rural areas while promoting sustainable use of natural and cultural resources.

Ultimately, agritourism development reflects a shift in rural development thinking from sectoral isolation toward integrated and place-based development approaches (Van Sandt et al., 2019). It positions rural communities not only as producers of agricultural commodities but also as active participants in service-based economies that combine agriculture, tourism and cultural heritage into a unified development pathway.

2.2 Conceptualizing Agritourism

Agritourism is conceptualized as a specialized form of tourism that takes place within working agricultural environments where visitors actively engage in farm-based experiences for educational, recreational, cultural, or consumption purposes (Barbieri et al., 2016). It involves direct interaction between tourists and agricultural production systems, allowing visitors to observe and participate in activities such as planting, harvesting, livestock rearing, and food processing. This immersive engagement distinguishes agritourism from passive forms of tourism by emphasizing experiential learning and hands-on participation (Wu et al., 2024). A key distinguishing feature of agritourism is its integration with active agricultural production systems (Ammirato et al., 2020). Unlike general rural tourism, which may occur in non-agricultural countryside settings, agritourism is inherently tied to functioning farms and agribusiness operations. These may include crop farms, plantations, livestock farms, and mixed farming systems that remain economically productive while simultaneously hosting visitors. Therefore, Qiu et al. (2021) proposed that this dual function creates a unique interaction between agricultural production and tourism services within the same spatial and economic setting.

Scholarly literature increasingly frames agritourism as a multifunctional farm enterprise that extends traditional agricultural value chains through the incorporation of tourism-related services (Eshun & Mensah, 2020). This multifunctionality allows farmers to diversify their income sources by combining primary production with value-added services such as guided farm tours, “pick-your-own” experiences, farm-gate sales, and cultural or educational demonstrations of rural practices. In doing so, agritourism transforms farms into hybrid enterprises that operate simultaneously as production units and service providers. This multifunctional nature enhances the economic utility of agricultural land by increasing its productivity beyond conventional farming outputs. It also strengthens consumer awareness of food systems by exposing visitors to the processes involved in agricultural production, processing, and distribution. As a result, agritourism plays an important role in reconnecting consumers with the origins of food while promoting appreciation for rural livelihoods and agricultural heritage (Pehin Dato Musa & Chin, 2022).

Consequently, Eshun & Mensah (2020) opined that agritourism serves both commercial and educational purposes. It positions farms as dual-purpose spaces of production and consumption, where economic exchange is combined with learning, leisure, and cultural interaction. This duality reinforces its significance as an innovative rural development strategy that contributes to both farm sustainability and rural economic diversification.

2.3 Agritourism as a Livelihood Diversification Strategy

A dominant theme in agritourism scholarship is its contribution to rural livelihood diversification, particularly in developing economies where agriculture remains the primary source of income for rural households (Cavalleri et al., 2022). In such contexts, farming activities are often characterized by high levels of uncertainty arising from seasonal production cycles, fluctuating commodity prices, limited access to markets, and increasing exposure to climate-related risks such as droughts and erratic rainfall. These constraints frequently result in unstable and insufficient household incomes, thereby necessitating alternative or supplementary livelihood strategies. Within this context, agritourism is increasingly adopted as a complementary income-generating strategy that helps stabilize rural livelihoods. By integrating tourism activities into agricultural systems, rural households are able to reduce overdependence on farm production alone and access additional revenue streams throughout the year (Neglo et al., 2021). This diversification is particularly important in regions where agricultural productivity is highly seasonal and vulnerable to external shocks, making income smoothing a critical development objective (Wu et al., 2024).

Empirical studies indicate that agritourism enhances livelihood diversification by enabling rural households to participate in a range of non-farm activities (Li et al., 2024; Zvavahera & Chigora, 2023). These include tour guiding, provision of accommodation and catering services, sale of farm produce and value-added goods, and participation in cultural or educational tourism experiences. Such activities expand household income portfolios while strengthening local value chains and increasing the economic circulation of resources within rural communities. Beyond its economic contributions, Sulaj & Themelko (2024) posited that agritourism also promotes social inclusion by creating opportunities for groups that are often marginalized in traditional agricultural systems, particularly women and young people. Their participation in agritourism-related enterprises enhances income equality and supports broader rural development goals. Consequently, Sarif & Charnnuwong (2024) opined that agritourism functions not only as an income diversification mechanism but also as a resilience-building strategy that strengthens household and community capacity to withstand economic, social, and

environmental uncertainties.

2.4 Livelihood Opportunities Generated Through Agritourism

Multiple and interrelated livelihood pathways associated with agritourism development have been identified, extending beyond direct income generation to broader socio-economic transformation in rural communities (Pehin Dato Musa & Chin, 2022). According to Cavalleri et al. (2022), agritourism creates a diversified livelihood structure that integrates employment, entrepreneurship, skills development, and community-level welfare improvements. These outcomes collectively contribute to strengthening rural economies and reducing vulnerability associated with reliance on subsistence agriculture. Direct employment opportunities constitute one of the most visible outcomes of agritourism development (Kipkorir et al., 2022). These opportunities commonly emerge in farm-based tourism operations such as tour guiding, interpretation services, hospitality provision, transportation services, and culinary activities linked to local food systems. Such employment is often locally embedded, thereby enabling rural residents to participate directly in tourism value chains without necessitating migration to urban centers.

Beyond employment, agritourism plays a significant role in stimulating micro and small enterprise development within rural settings (Bannor et al., 2022; Pehin Dato Musa & Chin, 2022). Increased tourist inflows generate demand for locally produced goods and services, encouraging entrepreneurial activities such as food vending, handicraft production, souvenir sales, accommodation services, and the marketing of agricultural products. These enterprises not only diversify household income sources but also strengthen local economic linkages and multiplier effects within rural economies. In addition, Eshun & Mensah (2020) postulated that agritourism is associated with capacity-building and human capital development. Training programs in modern agricultural practices, customer service, and hospitality management enhance the skills and productivity of rural residents. This improved human capital base increases the competitiveness of rural enterprises and supports long-term income generation potential.

At the community level, agritourism contributes to indirect livelihood improvements through investments in local infrastructure and social development initiatives (Bannor et al., 2022; Zvavahera & Chigora, 2023). These may include improved access to basic amenities such as water and sanitation, educational support schemes such as scholarships, and the development of small-scale community facilities. Collectively, these interventions enhance overall rural welfare and reinforce the broader development impact of agritourism.

2.5 Agritourism and Sustainable Rural Development

Beyond immediate livelihood outcomes, Baipai et al. (2023) and Garwi (2024) posited that agritourism is increasingly framed within sustainability discourse as a multidimensional development strategy that integrates economic viability, environmental stewardship, and socio-cultural preservation in rural areas. This perspective aligns with sustainable development principles that emphasize the need to balance resource use with long-term community wellbeing, particularly in rural regions where livelihoods are closely tied to natural ecosystems. From an environmental perspective, agritourism contributes to the preservation of agricultural landscapes by maintaining active farming systems that might otherwise be abandoned or converted to non-agricultural uses (Charles, 2020). The continued use of farmland for both production and tourism encourage more sustainable land management practices, including soil conservation, biodiversity protection, and reduced land degradation. In some cases, agritourism also promotes environmentally conscious farming practices as farmers seek to maintain aesthetically and ecologically attractive landscapes for visitors (Wu et al., 2024).

Socio-culturally, agritourism plays an important role in safeguarding rural heritage and reinforcing community identity (Garwi, 2024). It facilitates interaction between rural producers and visitors, enabling the transmission of traditional farming knowledge, local customs, and cultural practices. This interaction not only enhances visitor awareness of rural life but also strengthens community pride and cultural continuity within host communities. In this way, agritourism becomes a mechanism for cultural preservation and intercultural exchange. Economically, agritourism enhances rural resilience by reducing overdependence on a single livelihood source such as crop production (Bannor et al., 2022; Cavalleri et al., 2022). By integrating rural economies into broader tourism value chains, it diversifies income sources and improves household and community stability in the face of market and environmental shocks. The combined economic, environmental, and socio-cultural benefits position agritourism as a strategic instrument for sustainable rural transformation, particularly in settings characterized by poverty, limited diversification, and high agricultural dependence.

3. Methodology

This study adopted a pragmatic research paradigm and employed a case study design to examine the role of agritourism development in enhancing livelihood opportunities in rural communities. The case study approach

was considered appropriate because it enabled an in-depth investigation of agritourism within its real-life context in Adjeikrom, a pioneering agritourism destination in Ghana's Eastern Region. Guided by pragmatism, the study utilized a mixed-methods approach, integrating quantitative and qualitative techniques to provide a comprehensive understanding of the phenomenon. The convergent parallel design facilitated the simultaneous collection and analysis of both datasets, allowing for triangulation and validation of findings from multiple sources. Primary and secondary data were collected to address the study objectives.

Quantitative data were obtained through structured household questionnaires administered to 100 households selected using a multistage sampling procedure that combined cluster and simple random sampling techniques. In the first stage, the study area was divided into clusters based on communities actively engaged in agritourism activities. Selected communities constituted the primary sampling units. In the second stage, household lists were obtained with the assistance of community leaders, and households were selected through simple random sampling to ensure that each eligible household had an equal chance of participation. The sample comprised 100 households distributed across the selected communities. For the qualitative component, key informant interviews were conducted with tourism operators, community leaders, and officials from relevant institutions, while focus group discussions were held with residents involved in agritourism activities. This mixed-methods approach enabled the triangulation of quantitative and qualitative data to provide a comprehensive understanding of agritourism's contribution to rural livelihoods.

Qualitative data were gathered through in-depth interviews, focus group discussions, participant observation, and stakeholder consultations involving community members, tourism operators, local leaders, and representatives of the Ghana Tourism Authority. The inclusion of diverse stakeholder groups ensured the capture of multiple perspectives on agritourism development and its contribution to rural livelihoods. Data analysis involved both descriptive and inferential statistical techniques as well as thematic analysis. Quantitative data were coded, cleaned, and analyzed using IBM SPSS Statistics Version 24, generating correlation outputs to examine relationships among key variables. Qualitative data were transcribed, organized and analyzed thematically following established data reduction procedures to identify recurring patterns and insights. To enhance the validity, reliability and trustworthiness of the findings, the study employed methodological triangulation, pre-testing of research instruments, and rigorous data verification procedures. Ethical standards were maintained throughout the research process through informed consent, voluntary participation, confidentiality, and the protection of participants' identities.

4. Results

4.1 Agritourism Resources and Community Perceptions

The findings indicate that Adjeikrom possesses significant agritourism potential, anchored on three major attractions: cocoa farm tours, village tours, and mountain hiking activities. A large majority (94%) of respondents reported having visited these attractions, although only 14% had done so specifically for tourism purposes, largely due to their familiarity with the resources. Nevertheless, respondents unanimously identified these attractions as the primary factors drawing visitors to the community. Furthermore, 67% of respondents rated the agritourism resources as "very good," while 33% considered them "good," suggesting a strong local appreciation of the community's tourism assets.

Qualitative evidence further reinforced these findings. A 31-year-old female farmer noted that although residents are familiar with cocoa farming, greater awareness is needed regarding its tourism value and income-generating potential. Similarly, stakeholder interviews with tour guides and facility managers revealed that visitors are particularly attracted by opportunities to experience cocoa production processes, participate in harvesting activities, and interact with local culture. These findings suggest that the existing agritourism resources provide a strong foundation for local economic participation and livelihood enhancement.

4.2 Employment and Income-Generating Opportunities

Agritourism has created diverse livelihood opportunities for residents of Adjeikrom. Approximately 74% of respondents reported that their households benefit directly from agritourism activities through employment and business opportunities. These benefits include serving as tour guides, providing accommodation services, preparing meals for visitors, selling agricultural produce, and engaging in other tourism-related enterprises. Interviews with restaurant operators and guesthouse owners confirmed that tourist visits generate additional demand for local goods and services, thereby increasing household income and stimulating small business development within the community.

The study further examined the relationship between income level and livelihood improvement using Pearson's correlation analysis.

Livelihood improvement was operationalized as a composite measure derived from respondents' assessments

of changes in household welfare resulting from tourism activities. The construct was measured using indicators such as improvement in household income, access to education, healthcare, housing conditions, and overall living standards. Responses were captured on a five-point Likert scale and aggregated by calculating the mean score across the indicators. The internal consistency of the scale was assessed using Cronbach's alpha, which yielded a reliability coefficient of 0.88.

The results in Table 1 revealed a weak positive relationship ($r = 0.250, p = 0.080$) between respondents' income levels and perceived livelihood improvement. Although the relationship is relatively weak, it indicates that increases in income associated with agritourism activities contribute positively to household well-being. The coefficient of determination ($r^2 = 0.0625$) suggests that income accounts for approximately 6.25% of the variation in livelihood improvement, while other social and economic factors explain the remaining variation. These findings demonstrate that agritourism contributes to livelihood enhancement but operates alongside other determinants of rural welfare.

Table 1. Relationship between income level and livelihood improvement

Variable	Level of Income	Livelihood Improvement
Level of income	–	–
Pearson correlation	1	0.250
Significance (two-tailed)	–	0.080
<i>N</i>	100	100
Livelihood improvement	–	–
Pearson correlation	0.250	1
Significance (two-tailed)	0.080	–
<i>N</i>	100	100

Note: The en dash (–) denotes no data available.

4.3 Agricultural Capacity Building and Productivity Enhancement

Beyond direct employment, agritourism has contributed to strengthening agricultural livelihoods through training and technical support. The study found that 84% of respondents perceived an increase in cocoa production following the introduction of agritourism initiatives in the community. Respondents attributed this improvement to enhanced motivation among farmers, access to training on modern cocoa production techniques, and the provision of subsidized agricultural inputs such as fertilizers, pesticides, and agrochemicals.

Stakeholder interviews provided further insights into these benefits. Representatives from tourism-support organizations and facility managers explained that partnerships with organizations such as Cadbury Ghana, the Nature Conservation Research Center, Earthwatch Institute, and the Cocoa Research Institute of Ghana have enabled farmers to receive training in best agronomic practices, including disease management, weed control, harvesting techniques, and post-harvest handling. Approximately 60% of respondents acknowledged receiving such support. These interventions have enhanced agricultural productivity, increased the quality of cocoa production, and strengthened the economic resilience of farming households.

4.4 Community Development and Social Benefits

The development of agritourism has also generated broader community benefits. Respondents reported that tourism-related activities have contributed to the provision of community infrastructure and social support programs. Examples include the drilling of boreholes, opportunities for hiring chairs and mattresses during community events, and scholarship schemes sponsored by private organizations for academically talented but financially disadvantaged students. Interviews with local leaders and assembly representatives confirmed that agritourism has increased the visibility of the community and attracted external support from governmental and non-governmental organizations. These findings suggest that agritourism contributes not only to individual livelihoods but also to collective community development through investments in social infrastructure and human capital.

4.5 Agritourism Growth Potential and Sustainability of Livelihood Opportunities

The study further assessed residents' perceptions regarding the future potential of agritourism. Nearly all respondents (98%) believed that the community's agritourism resources have the capacity to attract more visitors. Although 44% noted that current visitation levels remain relatively low, respondents emphasized that the uniqueness of cocoa-based tourism, village experiences, and mountain hiking activities provides opportunities for future growth. Additionally, 78% reported frequent visits by international tourists, particularly from the United Kingdom, Germany, the Netherlands, and the United States.

As shown in Table 2, a Pearson correlation analysis revealed a weak but statistically significant positive relationship between income level and perceptions of agritourism development ($r = 0.340$, $p = 0.016$). This finding indicates that individuals with higher income levels tend to perceive agritourism more positively. Stakeholder interviews further revealed that year-round tourism activity, supported by hybrid cocoa varieties and diversified attractions, minimizes seasonality and creates more stable livelihood opportunities. Facility managers and tour guides emphasized that the availability of accommodation facilities and continuous visitor flows help sustain employment, local businesses, and income generation throughout the year. Collectively, these findings demonstrate that agritourism represents a viable pathway for enhancing livelihood opportunities and promoting sustainable rural development in Adjeikrom.

Table 2. Relationship between income level and agritourism perceived at Adjeikrom

Variable	Level of Income	Agritourism at Adjeikrom
Level of income	–	–
Pearson correlation	1	0.340*
Significance (two-tailed)	–	0.016
<i>N</i>	100	100
Agritourism at Adjeikrom	–	–
Pearson correlation	0.340*	1
Significance (two-tailed)	0.016	–
<i>N</i>	100	100

Note: The en dash (–) denotes no data available; * means correlation is significant at the 0.05 level (two-tailed).

5. Discussion of Findings

The findings demonstrate that Adjeikrom possesses considerable agritourism potential through its cocoa farm tours, village tours, and mountain hiking activities, which are widely recognized by residents as the major attractions drawing visitors to the community. This finding supports the conceptualization of agritourism by Ammirato et al. (2020), Barbieri et al. (2016), and Eshun & Mensah (2020), who argued that agritourism is rooted in active agricultural environments where visitors engage directly with farming activities and rural life. The opportunities for tourists to observe cocoa production, participate in harvesting activities, and interact with local culture reflect the experiential and educational nature of agritourism emphasized by Wu et al. (2024). The positive assessment of the community's agritourism resources further confirms the argument by Kothari & Perwej (2021) and Streifeneder et al. (2023) that agricultural resources can be transformed into tourism assets that generate economic and social benefits for rural communities. However, the finding that only a few residents visited these attractions specifically for tourism purposes suggests that familiarity with local resources may limit residents' appreciation of their tourism value, indicating the need for greater awareness creation on the economic potential of agritourism.

The study further revealed that agritourism has created diverse livelihood opportunities for residents through employment and business activities such as tour guiding, accommodation services, food preparation, produce sales, and other tourism-related enterprises. These findings are consistent with the livelihood diversification perspective advanced by Cavalleri et al. (2022), Neglo et al., (2021), and Zvavahera & Chigora (2023), who argued that agritourism provides alternative income opportunities that reduce dependence on traditional farming. The increased demand for local goods and services reported by restaurant operators and guesthouse owners also supports the assertions of Bannor et al. (2022) and Pehin Dato Musa & Chin (2022) that agritourism stimulates entrepreneurship and strengthens local economic linkages.

The correlation analysis revealed a weak positive relationship between income level and livelihood improvement ($r = 0.250$, $p = 0.080$). However, the relationship was not statistically significant at the 5% significance level, suggesting that the observed association may have occurred by chance. Therefore, while respondents who reported higher income levels from agritourism activities tended to report slightly better livelihood outcomes, the findings do not provide sufficient statistical evidence to conclude that income derived from agritourism significantly influences livelihood improvement. The coefficient of determination ($r^2 = 0.0625$) further indicates that income accounts for only 6.25% of the variation in livelihood improvement, implying that a substantial proportion of livelihood outcomes is explained by other factors. These findings suggest that agritourism may contribute to rural livelihoods, but its influence should be viewed as one of several factors affecting household welfare rather than the primary determinant. This finding is consistent with that of Ma et al. (2024), who argued that tourism contributes to rural development as part of a broader livelihood system that includes social, economic, institutional, and environmental factors.

Another important finding is that agritourism has contributed to agricultural capacity building and community development. The reported increase in cocoa production and the training received by farmers support the views of Cavalleri et al. (2022) and Eshun & Mensah (2020), who argued that agritourism promotes skills development

and enhances rural human capital. Through partnerships with organizations such as Cadbury Ghana, the Nature Conservation Research Center, Earthwatch Institute, and the Cocoa Research Institute of Ghana, farmers have gained knowledge in disease management, harvesting techniques, and post-harvest handling practices, leading to improved productivity. These findings demonstrate that agritourism extends beyond tourism activities to support agricultural development and strengthen rural livelihoods. Similarly, the provision of boreholes, scholarship schemes, and other community support initiatives agrees with the findings of Bannor et al. (2022) and Zvavahera & Chigora (2023), who emphasized that agritourism contributes to community welfare through investments in infrastructure and social development. The findings therefore support the view that agritourism generates both individual and collective benefits that enhance rural development.

Finally, the findings indicate strong prospects for the future growth and sustainability of agritourism in Adjeikrom. The widespread belief among respondents that the community has the capacity to attract more visitors supports the argument by Unjia et al. (2024) that agritourism enables rural communities to diversify their economies and strengthen market-oriented livelihood systems. The significant positive relationship between income level and perceptions of agritourism development ($r = 0.340$, $p = 0.016$) further suggests that residents who benefit economically from tourism tend to have more favorable perceptions of its development. The continuous flow of both domestic and international visitors and the availability of diverse attractions throughout the year also support the arguments of Bannor et al. (2022), Garwi (2024), and Zvavahera & Chigora (2023) that agritourism contributes to sustainable rural development by generating stable livelihood opportunities, promoting community resilience, and transforming rural spaces into multifunctional environments that combine production, tourism, and cultural exchange. Overall, the findings strongly agree with the reviewed literature and demonstrate that agritourism serves as an important pathway for livelihood improvement and sustainable rural development in Adjeikrom.

Although agritourism contributes to income generation and livelihood improvement, several challenges may limit its long-term benefits. Field observations revealed that inadequate infrastructure, particularly poor road networks, affects accessibility to agritourism sites. Seasonal fluctuations in tourist arrivals also result in inconsistent income for participating households. Additionally, limited financial resources, technical skills and awareness constrain the participation of some community members. Weak coordination among tourism agencies, agricultural institutions and local authorities further hindered effective planning and management. Addressing these challenges is essential for enhancing the sustainability and livelihood impacts of agritourism.

6. Conclusions

This study concludes that agritourism serves as a viable pathway for rural livelihood enhancement in Adjeikrom, Ghana, by integrating agriculture with tourism to generate income, employment, and broader community development benefits. The findings show that the community possesses strong agritourism resources such as cocoa farm tours, village experiences, and mountain hiking which are widely recognized and positively rated by residents, forming a solid foundation for tourism development. Although most residents are familiar with these attractions, a smaller proportion engages with them for tourism purposes, indicating that agritourism is still emerging in terms of structured economic use. Nonetheless, these resources have begun to support household livelihoods through direct and indirect employment in guiding, hospitality, food services, and small-scale trade, while also stimulating local business activities.

The study found a weak positive relationship between income level and livelihood improvement among households engaged in agritourism activities. However, the relationship was not statistically significant ($p > 0.05$), indicating that the evidence is insufficient to establish a significant association between income and livelihood improvement within the study area. While agritourism may contribute to household welfare through income generation, the findings suggest that livelihood improvement is influenced by a range of factors beyond income alone. Consequently, agritourism should be regarded as a complementary contributor to rural livelihoods rather than a primary explanatory factor of livelihood improvement based on the current analysis. In addition, agritourism has strengthened agricultural productivity through training, input support, and institutional partnerships, while also contributing to community development through infrastructure and social support initiatives. The high level of optimism among residents regarding future growth, coupled with increasing tourist visits, further confirms the sustainability potential of agritourism in the area. Overall, the study demonstrates that agritourism in Adjeikrom is gradually evolving into an important rural development strategy that enhances livelihoods through multiple economic and social pathways; therefore, it is recommended that relevant stakeholders, including the Ghana Tourism Authority and local government authorities, intensify investment in infrastructure, marketing, and community capacity-building to strengthen agritourism development, while also formalizing community-based participation structures to ensure inclusive and sustainable benefits for rural households.

Author Contributions

Conceptualization, A.K., F.K.M., and K.N.; methodology, K.N.; software, F.K.M.; validation, A.K., K.N., and F.K.M.; formal analysis, F.K.M.; investigation, F.K.M.; resources, A.K., K.N., and F.K.M.; data curation, F.K.M.; writing original draft preparation, A.K.; writing review and editing, A.K. and F.K.M.; visualization, F.K.M.; supervision, A.K.; project administration, A.K., K.N., and F.K.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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